

Talgat Zharkynbay

“Module 2 Homework”

Load packages

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 3.5.3
```

```
library(sandwich)
```

```
## Warning: package 'sandwich' was built under R version 3.5.3
```

```
library(car)
```

```
## Warning: package 'car' was built under R version 3.5.3
```

```
## Warning: package 'carData' was built under R version 3.5.2
```

```
library(lmtest)
```

```
## Warning: package 'lmtest' was built under R version 3.5.3
```

```
## Warning: package 'zoo' was built under R version 3.5.3
```

```
library(lfe)
```

```
## Warning: package 'lfe' was built under R version 3.5.3
```

Part A

Q1

i)

```
Signals=read.csv(file="Signals_A_62016.csv", header=TRUE, as.is = TRUE, na.strings = c("NA",  
".", ""))  
SignalsDF_1992 = Signals[Signals$PFormDate == "30/06/1992", ]  
#Creating the Equation Model 1  
EQ1=lm(AvgMonRet~log(Size)+log(BTM)+ROE+NSI+NXF, data = SignalsDF_1992)  
SummaryEQ1=summary(EQ1)  
SummaryEQ1$coefficients
```

```
##              Estimate   Std. Error   t value   Pr(>|t|)
## (Intercept)  4.073974e+00 0.2877517538 14.1579467 2.613563e-43
## log(Size)    -2.896967e-01 0.0540371490 -5.3610657 9.295732e-08
## log(BTM)     6.250812e-01 0.1122507891  5.5686127 2.938629e-08
## ROE          -6.764786e-02 0.1770648831 -0.3820512 7.024666e-01
## NSI          -9.613134e-01 0.4754715868 -2.0218104 4.333773e-02
## NXF          -6.674025e-05 0.0002970645 -0.2246659 8.222637e-01
```

A 1% increase in Size decreases the Average Returns by -0.00290. A 1% increase in BTM increases the Average Returns by 0.00625. A 1 unit increase in ROE decreases the Average Returns by 0.068. A 1 unit increase in NSI decreases the Average Returns by 0.961. A 1 unit increase in NXF decreases the Average Returns by -0.0001. The formula for $\hat{b}_j$ is $\hat{b}_j + \sum V_{i,j}(\hat{b}_j) U_i / \sum V^2_{i,j}(\hat{b}_j)$ The formula for b_0 is $b_0 + (b_1 - b_1(\hat{b}_1))x_1(\bar{x}) + \dots + (b_p - b_p(\hat{b}_p))x_p(\bar{x}) + U(\hat{b})$

ii)

```
SummaryEQ1$coefficients
```

```
##              Estimate   Std. Error   t value   Pr(>|t|)
## (Intercept)  4.073974e+00 0.2877517538 14.1579467 2.613563e-43
## log(Size)    -2.896967e-01 0.0540371490 -5.3610657 9.295732e-08
## log(BTM)     6.250812e-01 0.1122507891  5.5686127 2.938629e-08
## ROE          -6.764786e-02 0.1770648831 -0.3820512 7.024666e-01
## NSI          -9.613134e-01 0.4754715868 -2.0218104 4.333773e-02
## NXF          -6.674025e-05 0.0002970645 -0.2246659 8.222637e-01
```

ROE and NXF have large standard errors, and hence we can not reject the null hypothesis that b_3 and b_4 are equal to zero. The formula for standard error is $\sqrt{\sum(Y - \hat{Y})^2 / n}$.

iii)

```
SummaryEQ1$coefficients
```

```
##              Estimate   Std. Error   t value   Pr(>|t|)
## (Intercept)  4.073974e+00 0.2877517538 14.1579467 2.613563e-43
## log(Size)    -2.896967e-01 0.0540371490 -5.3610657 9.295732e-08
## log(BTM)     6.250812e-01 0.1122507891  5.5686127 2.938629e-08
## ROE          -6.764786e-02 0.1770648831 -0.3820512 7.024666e-01
## NSI          -9.613134e-01 0.4754715868 -2.0218104 4.333773e-02
## NXF          -6.674025e-05 0.0002970645 -0.2246659 8.222637e-01
```

Coefficients with large SE's have smaller t-values. Hence, ROE and NXF have smaller t-values. Formula for t-value is $(\hat{b}_j - b_j) / se(\hat{b}_j)$.

iv)

```
SummaryEQ1$sigma
```

```
## [1] 4.352023
```

The residual standard error is calculated as $\sqrt{SSR / (n - p - 1)}$, or it is the square root of the Mean Squared Error.

v)

SummaryEQ1\$r.squared

[1] 0.04387335

SummaryEQ1\$adj.r.squared

[1] 0.04132368

R^2 and adjusted R^2 are measurements of Goodness-Of-Fit for the MLR. $R^2 = \text{SSE}/\text{SST}$, while $R_a^2 = 1 - (n-1) * (1 - R^2) / (n-p-1)$

vi)

SummaryEQ1\$fstatistic

##	value	numdf	dendf
##	17.20745	5.00000	1875.00000

F-statistic checks if the MLR model's coefficients are significant overall. The formula is $((\text{SSR} - \text{SSR}_r) / q) / (\text{SSR}_r / (n - p - 1))$. Our model has high F-stat and hence we reject the null hypothesis that all coefficients are zero.

Q2

```
WhiteTestDF=data.frame(UhatSQ=SummaryEQ1$residuals^2,Yhat=fitted(EQ1),
                        YhatSQ=fitted(EQ1)^2)

Test=lm(UhatSQ~Yhat+YhatSQ, data = WhiteTestDF)

vcovHC0=vcovHC(Test, type = "HC0")

linearHypothesis(Test, c("Yhat=0", "YhatSQ=0"), vcov. =vcovHC0, test = "F")
```

```
## Linear hypothesis test
##
## Hypothesis:
## Yhat = 0
## YhatSQ = 0
##
## Model 1: restricted model
## Model 2: UhatSQ ~ Yhat + YhatSQ
##
## Note: Coefficient covariance matrix supplied.
##
##   Res.Df Df      F Pr(>F)
## 1    1880
## 2    1878  2 2.6107 0.07375 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The p-value is above the chosen significance level of 2%, hence we accept homoskedasticity.

Q3

i)

```
confint(EQ1, "log(Size)", level = 0.95, alternative="two.sided")
```

```
##           2.5 %      97.5 %
## log(Size) -0.395676 -0.1837174
```

```
confint(EQ1, "log(BTM)", level = 0.95, alternative="two.sided")
```

```
##           2.5 %      97.5 %
## log(BTM) 0.4049316 0.8452308
```

```
confint(EQ1, "ROE", level = 0.95, alternative="two.sided")
```

```
##           2.5 %      97.5 %
## ROE -0.4149128 0.2796171
```

```
confint(EQ1, "NSI", level = 0.95, alternative="two.sided")
```

```
##           2.5 %      97.5 %
## NSI -1.893823 -0.02880426
```

```
confint(EQ1, "NXF", level = 0.95, alternative="two.sided")
```

```
##           2.5 %      97.5 %
## NXF -0.0006493521 0.0005158715
```

#When there is homoskedasticity, we can use OLS SE's with confint() function.

ii)

```
newDF=data.frame(Size=quantile(SignalsDF_1992$Size, prob=0.2, na.rm=T),
                  NSI=quantile(SignalsDF_1992$NSI, prob=0.2, na.rm=T),
                  NXF=quantile(SignalsDF_1992$NXF, prob=0.2, na.rm=T),
                  BTM=quantile(SignalsDF_1992$BTM, prob=0.8, na.rm=T),
                  ROE=quantile(SignalsDF_1992$ROE, prob=0.5, na.rm=T))

predict(EQ1, newDF)
```

```
##      20%
## 3.21307
```

```
# This is a MEAN/EXPECTED value of Average Monthly Returns. Now we need a CI:
```

```
SignalsDF_1992$Size_New=SignalsDF_1992$Size-quantile(SignalsDF_1992$Size, prob=0.2, na.rm=T)
SignalsDF_1992$BTM_New=SignalsDF_1992$BTM-quantile(SignalsDF_1992$BTM, prob=0.8, na.rm=T)
SignalsDF_1992$ROE_New=SignalsDF_1992$ROE-quantile(SignalsDF_1992$ROE, prob=0.5, na.rm=T)
SignalsDF_1992$NSI_New=SignalsDF_1992$NSI-quantile(SignalsDF_1992$NSI, prob=0.2, na.rm=T)
SignalsDF_1992$NXF_New=SignalsDF_1992$NXF-quantile(SignalsDF_1992$NXF, prob=0.2, na.rm=T)
```

```
EQ1_New=lm(AvgMonRet~log(Size_New)+log(BTM_New)+ROE_New+NSI_New+NXF_New,
           data = SignalsDF_1992)
```

```
## Warning in log(Size_New): созданы NaN
```

```
## Warning in log(BTM_New): созданы NaN
```

```
SummaryEQ1_New=summary(EQ1_New)
```

```
confint(EQ1_New, "(Intercept)", level = 0.95, alternative="two.sided")
```

```
##                2.5 %    97.5 %
## (Intercept) 2.009811 4.553812
```

```
#This is Interval estimate for predicted mean of AvgMonRet. Now we can do
#the Interval estimate for the specific value:
```

```
SE_Spc=sqrt(vcov(EQ1_New)[1,1]+SummaryEQ1$sigma^2)
SE_Spc
```

```
## [1] 4.399607
```

```
EQ1_New$coefficients["(Intercept)"]-qt(p=0.975, df=1875)*SE_Spc
```

```
## (Intercept)
##          -5.34683
```

```
EQ1_New$coefficients["(Intercept)"]-qt(p=0.025, df=1875)*SE_Spc
```

```
## (Intercept)
##          11.91045
```

```
#The interval estimate for the specific value is [-5.34683, 11.91045]
```

iii)

```
# We can use the F-Test with the linearhypothesis function
linearHypothesis(EQ1, c("ROE=0", "NSI=0", "NXF=0"),test = "F" )
```

```
## Linear hypothesis test
##
## Hypothesis:
## ROE = 0
## NSI = 0
## NXF = 0
##
## Model 1: restricted model
## Model 2: AvgMonRet ~ log(Size) + log(BTM) + ROE + NSI + NXF
##
##   Res.Df  RSS Df Sum of Sq    F Pr(>F)
## 1   1878 35593
## 2   1875 35513   3    80.684 1.42 0.2351
```

We have a 5% significance level and p-value of 0.2351, which is much higher. Hence, we accept H_0 and conclude that b_3, b_4 and b_5 are not significantly different from zero.

iv)

```
linearHypothesis(EQ1, "NSI-3*log(Size)=0")
```

```
## Linear hypothesis test
##
## Hypothesis:
## - 3 log(Size) + NSI = 0
##
## Model 1: restricted model
## Model 2: AvgMonRet ~ log(Size) + log(BTM) + ROE + NSI + NXF
##
##   Res.Df  RSS Df Sum of Sq    F Pr(>F)
## 1   1876 35513
## 2   1875 35513   1    0.67196 0.0355 0.8506
```

Again we have very high p-value, hence we fail to reject the H_0 and we conclude that in the population of interest the partial effect of NSI on future returns is 3 times as big as the partial effect of $\log(\text{Size})$

v)

```
linearHypothesis(EQ1, c("ROE=0", "NXF=0", "NSI-3*log(Size)=0"),
                  test = "F" )
```

```
## Linear hypothesis test
##
## Hypothesis:
## ROE = 0
## NXF = 0
## - 3 log(Size) + NSI = 0
##
## Model 1: restricted model
## Model 2: AvgMonRet ~ log(Size) + log(BTM) + ROE + NSI + NXF
##
##      Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1      1878 35517
## 2      1875 35513   3    4.2802 0.0753 0.9733
```

I used the F-test. Again at the p-value of 0.9733 we fail to reject H_0 , and we conclude that statement is true.

Q4)

```
bptest(EQ1)
```

```
##
## studentized Breusch-Pagan test
##
## data: EQ1
## BP = 11.729, df = 5, p-value = 0.0387
```

#The p-value is below the significance level of 7%. Hence, we reject the H_0 of homoskedasticity.

Q5)

We will now use the Robust interval estimates with `coefci` function ### i)

```
vcovHC0=vcovHC(EQ1, type = "HC0")
coefci(EQ1, "log(Size)", level = 0.95, vcov. = vcovHC0)
```

```
##              2.5 %      97.5 %
## log(Size) -0.4161228 -0.1632706
```

```
coefci(EQ1, "log(BTM)", level = 0.95, vcov. = vcovHC0)
```

```
##              2.5 %      97.5 %
## log(BTM) 0.4192095 0.8309528
```

```
coefci(EQ1, "ROE", level = 0.95, vcov. = vcovHC0)
```

```
##              2.5 %      97.5 %
## ROE -0.3150521 0.1797564
```

```
coefci(EQ1, "NSI", level = 0.95, vcov. = vcovHC0)
```

```
##          2.5 %      97.5 %
## NSI -1.860533 -0.06209397
```

```
coefci(EQ1, "NXF", level = 0.95, vcov. = vcovHC0)
```

```
##          2.5 %      97.5 %
## NXF -0.0003810501 0.0002475696
```

*#Since we rejected homoskedasticity in Q4, we used Robust Standard Errors and
#received robust confidence intervals for our coefficients.*

ii)

```
newDF=data.frame(Size=quantile(SignalsDF_1992$Size, prob=0.2, na.rm=T),
                  NSI=quantile(SignalsDF_1992$NSI, prob=0.2, na.rm=T),
                  NXF=quantile(SignalsDF_1992$NXF, prob=0.2, na.rm=T),
                  BTM=quantile(SignalsDF_1992$BTM, prob=0.8, na.rm=T),
                  ROE=quantile(SignalsDF_1992$ROE, prob=0.5, na.rm=T))

predict(EQ1, newDF)
```

```
##      20%
## 3.21307
```

This is a MEAN/EXPECTED value of Average Monthly Returns. Now we need a CI:

```
SignalsDF_1992$Size_New=SignalsDF_1992$Size-quantile(SignalsDF_1992$Size, prob=0.2, na.rm=T)
SignalsDF_1992$BTM_New=SignalsDF_1992$BTM-quantile(SignalsDF_1992$BTM, prob=0.8, na.rm=T)
SignalsDF_1992$ROE_New=SignalsDF_1992$ROE-quantile(SignalsDF_1992$ROE, prob=0.5, na.rm=T)
SignalsDF_1992$NSI_New=SignalsDF_1992$NSI-quantile(SignalsDF_1992$NSI, prob=0.2, na.rm=T)
SignalsDF_1992$NXF_New=SignalsDF_1992$NXF-quantile(SignalsDF_1992$NXF, prob=0.2, na.rm=T)

EQ1_New=lm(AvgMonRet~log(Size_New)+log(BTM_New)+ROE_New+NSI_New+NXF_New,
           data = SignalsDF_1992)
```

```
## Warning in log(Size_New): созданы NaN
```

```
## Warning in log(BTM_New): созданы NaN
```

```
SummaryEQ1_New=summary(EQ1_New)

coefci(EQ1_New, "(Intercept)", level = 0.95, alternative="two.sided")
```

```
##          2.5 %      97.5 %
## (Intercept) 2.009811 4.553812
```



```
#This is Interval estimate for predicted mean of AvgMonRet. Now we can do
#the Interval estimate for the specific value:
SE_Spc=sqrt(vcov(EQ1_New)[1,1]+SummaryEQ1$sigma^2)
SE_Spc
```

```
## [1] 4.399607
```

```
EQ1_New$coefficients["(Intercept)"]-qt(p=0.975, df=1875)*SE_Spc
```

```
## (Intercept)
##      -5.34683
```

```
EQ1_New$coefficients["(Intercept)"]-qt(p=0.025, df=1875)*SE_Spc
```

```
## (Intercept)
##      11.91045
```

```
#The interval estimate for the specific value is [-5.34683, 11.91045]
#This results are the same in Q3 with homoskedasticity, because OLS method
#point estimators are still unbiased.
```

iii)

```
# We can use the F-Test with the linearhypothesis function
linearHypothesis(EQ1, c("ROE=0", "NSI=0", "NXF=0"),
                  vcov. =vcovHC0, test = "F" )
```

```
## Linear hypothesis test
##
## Hypothesis:
## ROE = 0
## NSI = 0
## NXF = 0
##
## Model 1: restricted model
## Model 2: AvgMonRet ~ log(Size) + log(BTM) + ROE + NSI + NXF
##
## Note: Coefficient covariance matrix supplied.
##
##   Res.Df Df    F Pr(>F)
## 1    1878
## 2    1875  3 1.5489    0.2
```

```
#We have a 5% significance level and p-value of 0.2, which is much higher.
#Hence, we accept H0 and conclude that b3,b4 and b5 are not significantly
#different from zero. With OLS SE's we had p-value of 0.2351.
```

iv)

```
linearHypothesis(EQ1, "NSI-3*log(Size)=0",vcov. =vcovHC0)
```

```
## Linear hypothesis test
##
## Hypothesis:
## - 3 log(Size) + NSI = 0
##
## Model 1: restricted model
## Model 2: AvgMonRet ~ log(Size) + log(BTM) + ROE + NSI + NXF
##
## Note: Coefficient covariance matrix supplied.
##
##   Res.Df Df      F Pr(>F)
## 1    1876
## 2    1875  1 0.0413  0.839
```

#Again we have very high p-value of 0.84, hence we fail to reject the H_0 and we conclude that in the population of interest the partial effect of NSI on future returns is 3 times as big as the partial effect of $\log(\text{Size})$. With OLS SE's we had p-value of 0.8506.

v)

```
linearHypothesis(EQ1, c("ROE=0", "NXF=0", "NSI-3*log(Size)=0"),
                 vcov. =vcovHC0, test = "F" )
```

```
## Linear hypothesis test
##
## Hypothesis:
## ROE = 0
## NXF = 0
## - 3 log(Size) + NSI = 0
##
## Model 1: restricted model
## Model 2: AvgMonRet ~ log(Size) + log(BTM) + ROE + NSI + NXF
##
## Note: Coefficient covariance matrix supplied.
##
##   Res.Df Df      F Pr(>F)
## 1    1878
## 2    1875  3 0.1614 0.9223
```

#I used the F-test. Again at the p-value of 0.9223 we fail to reject H_0 , and we conclude that statement is true. With OLS SE's we had p-value of 0.9733

Q6)

i)

```
EQ1_CL=felm(AvgMonRet~log(Size)+log(BTM)+ROE+NSI+NXF |0|0| BBHL,
            data = SignalsDF_1992)
SummaryEQ1_CL=summary(EQ1_CL)
SummaryEQ1_CL$coefficients
```

```
##              Estimate Cluster s.e.      t value      Pr(>|t|)
## (Intercept)  4.073974e+00 0.4563664463  8.9269797 1.018360e-18
## log(Size)    -2.896967e-01 0.0764391580 -3.7898992 1.554689e-04
## log(BTM)     6.250812e-01 0.1343619446  4.6522188 3.513561e-06
## ROE          -6.764786e-02 0.1153889856 -0.5862592 5.577719e-01
## NSI          -9.613134e-01 0.5545213976 -1.7335912 8.315503e-02
## NXF          -6.674025e-05 0.0001542868 -0.4325727 6.653750e-01
```

*#We found Cluster Robust Standard Errors and corresponding t and p values
 #Such technique should be used in financial data sets, because very often
 #there is not only heteroskedasticity in the data, but also cross-correlation.*

```
SummaryEQ1_CL$coefficients
```

```
##              Estimate Std. Error      t value      Pr(>|t|)
## (Intercept)  4.073974e+00 0.2877517538 14.1579467 2.613563e-43
## log(Size)    -2.896967e-01 0.0540371490 -5.3610657 9.295732e-08
## log(BTM)     6.250812e-01 0.1122507891  5.5686127 2.938629e-08
## ROE          -6.764786e-02 0.1770648831 -0.3820512 7.024666e-01
## NSI          -9.613134e-01 0.4754715868 -2.0218104 4.333773e-02
## NXF          -6.674025e-05 0.0002970645 -0.2246659 8.222637e-01
```

*#The results are different from the results of Question 3. Cluster SE's are
 #typically larger and t-values are smaller for most of estimates.*

ii)

```
vcovCluster=EQ1_CL$clustervcv

linearHypothesis(EQ1_CL, c("ROE=0", "NSI=0", "NXF=0"),
                 vcov. =vcovCluster, test = "F" )
```

```
## Linear hypothesis test
##
## Hypothesis:
## ROE = 0
## NSI = 0
## NXF = 0
##
## Model 1: restricted model
## Model 2: AvgMonRet ~ log(Size) + log(BTM) + ROE + NSI + NXF | 0 | 0 |
##      BBHL
##
## Note: Coefficient covariance matrix supplied.
##
##   Res.Df Df      F Pr(>F)
## 1    1878
## 2    1875  3 1.5301 0.2047
```

```
#We generated var-covar matrix with cluster-robust estimates and used that
#matrix in the F-test. As a result, we have a p-value of 0.2047. Hence,
#we can not reject the H0.
```

###Part B

Q1)

i)

```
EQ2=lm(AvgMonRet~log(Size)+log(BTM)+ROE+NSI+NXF, data = Signals)
SummaryEQ2=summary(EQ2)
SummaryEQ2$coefficients
```

##		Estimate	Std. Error	t value	Pr(> t)
##	(Intercept)	3.317109e+00	0.0471461025	70.358067	0.000000e+00
##	log(Size)	-2.543720e-01	0.0084849219	-29.979301	1.857746e-196
##	log(BTM)	3.171992e-01	0.0197253279	16.080807	4.222881e-58
##	ROE	-4.646452e-03	0.0037670512	-1.233445	2.174130e-01
##	NSI	-2.672897e-01	0.0769428186	-3.473875	5.132500e-04
##	NXF	-7.605044e-05	0.0000192296	-3.954864	7.663898e-05

```
#A 1% increase in Size decreases the Average Returns by 0.0025473.
#A 1% increase in BTM increases the Average Returns by 0.00317.
#A 1 unit increase in ROE decreases the Average Returns by 0.00464.
#A 1 unit increase in NSI decreases the Average Returns by 0.267.
#A 1 unit increase in NXF decreases the Average Returns by 0.00076,
```

ii)

```
SummaryEQ2$coefficients
```

##		Estimate	Std. Error	t value	Pr(> t)
##	(Intercept)	3.317109e+00	0.0471461025	70.358067	0.000000e+00
##	log(Size)	-2.543720e-01	0.0084849219	-29.979301	1.857746e-196
##	log(BTM)	3.171992e-01	0.0197253279	16.080807	4.222881e-58
##	ROE	-4.646452e-03	0.0037670512	-1.233445	2.174130e-01
##	NSI	-2.672897e-01	0.0769428186	-3.473875	5.132500e-04
##	NXF	-7.605044e-05	0.0000192296	-3.954864	7.663898e-05

```
#ROE has large standard error, and hence we can not reject
#the null hypothesis that b3 is equal to zero. Other estimates have lower SE's
#hence they are significantly different from zero.
```

iii)

```
SummaryEQ2$coefficients
```

```
##           Estimate   Std. Error   t value   Pr(>|t|)
## (Intercept)  3.317109e+00  0.0471461025  70.358067  0.000000e+00
## log(Size)    -2.543720e-01  0.0084849219 -29.979301  1.857746e-196
## log(BTM)     3.171992e-01  0.0197253279  16.080807  4.222881e-58
## ROE         -4.646452e-03  0.0037670512  -1.233445  2.174130e-01
## NSI         -2.672897e-01  0.0769428186  -3.473875  5.132500e-04
## NXF         -7.605044e-05  0.0000192296  -3.954864  7.663898e-05
```

*#Coefficients with large SE's have smaller t-values. Hence, ROE has
#smaller t-value of 1.233. Size and BTM have very high t-values.*

iv)

```
SummaryEQ2$sigma
```

```
## [1] 4.809708
```

*# The residual standard error is calculated as $\sqrt{SSR/n-p-1}$, or it is
the square root of the Mean Squared Error.*

v)

```
SummaryEQ2$r.squared
```

```
## [1] 0.01960264
```

```
SummaryEQ2$adj.r.squared
```

```
## [1] 0.01954608
```

*#R² and adjusted R² are measurements of Goodness-Of-Fit for the MLR.
#R²=SSE/SST, while $Ra^2=1-(n-1)*(1-R^2)/(n-p-1)$*

vi)

```
SummaryEQ2$fstatistic
```

```
##      value      numdf      dendf
##  346.5621    5.0000  86664.0000
```

*#F-statistic checks if the MLR model's coefficients are significant overall.
#The formula is $((SSRr-SSRur)/q)/(SSRur/n-p-1)$. Our model has high F-stat and
#hence we reject the null hypothesis that all coefficients are zero.*

Q2)

SummaryEQ1

```
##
## Call:
## lm(formula = AvgMonRet ~ log(Size) + log(BTM) + ROE + NSI + NXF,
##     data = SignalsDF_1992)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.944  -2.107  -0.298   1.595  75.162
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.074e+00  2.878e-01  14.158  < 2e-16 ***
## log(Size)    -2.897e-01  5.404e-02  -5.361  9.30e-08 ***
## log(BTM)     6.251e-01  1.123e-01   5.569  2.94e-08 ***
## ROE          -6.765e-02  1.771e-01  -0.382   0.7025
## NSI          -9.613e-01  4.755e-01  -2.022   0.0433 *
## NXF          -6.674e-05  2.971e-04  -0.225   0.8223
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.352 on 1875 degrees of freedom
## (612 observations deleted due to missingness)
## Multiple R-squared:  0.04387, Adjusted R-squared:  0.04132
## F-statistic: 17.21 on 5 and 1875 DF, p-value: < 2.2e-16
```

SummaryEQ2

```
##
## Call:
## lm(formula = AvgMonRet ~ log(Size) + log(BTM) + ROE + NSI + NXF,
##     data = Signals)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -72.647  -2.343  -0.280   1.881 206.871
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.317e+00  4.715e-02  70.358  < 2e-16 ***
## log(Size)    -2.544e-01  8.485e-03 -29.979  < 2e-16 ***
## log(BTM)     3.172e-01  1.973e-02  16.081  < 2e-16 ***
## ROE          -4.646e-03  3.767e-03  -1.233  0.217413
## NSI          -2.673e-01  7.694e-02  -3.474  0.000513 ***
## NXF          -7.605e-05  1.923e-05  -3.955  7.66e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.81 on 86664 degrees of freedom
## (32793 observations deleted due to missingness)
## Multiple R-squared:  0.0196, Adjusted R-squared:  0.01955
## F-statistic: 346.6 on 5 and 86664 DF, p-value: < 2.2e-16
```

Previously, both NXF and ROE were insignificant, in the second model only ROE is statistically insignificant. In both models estimates have the same signs and hence the same effects. Also, estimates in the second model are smaller than in the first model. Second model has lower R^2 and adjusted R^2 compared to the first model.

Q3)

i)

```
EQ2_CL_Perm=felm(AvgMonRet~log(Size)+log(BTM)+ROE+NSI+NXF |0|0| PermNo,
                  data = Signals)
SummaryEQ2_CL_Perm=summary(EQ2_CL_Perm)
SummaryEQ2_CL_Perm$coefficients
```

##		Estimate	Cluster s.e.	t value	Pr(> t)
##	(Intercept)	3.317109e+00	5.482327e-02	60.505491	0.000000e+00
##	log(Size)	-2.543720e-01	8.945256e-03	-28.436527	4.683708e-177
##	log(BTM)	3.171992e-01	2.247027e-02	14.116392	3.378759e-45
##	ROE	-4.646452e-03	4.574835e-03	-1.015655	3.097968e-01
##	NSI	-2.672897e-01	9.181051e-02	-2.911320	3.599972e-03
##	NXF	-7.605044e-05	1.867931e-05	-4.071374	4.677815e-05

```
SummaryEQ2$coefficients
```

##		Estimate	Std. Error	t value	Pr(> t)
##	(Intercept)	3.317109e+00	0.0471461025	70.358067	0.000000e+00
##	log(Size)	-2.543720e-01	0.0084849219	-29.979301	1.857746e-196
##	log(BTM)	3.171992e-01	0.0197253279	16.080807	4.222881e-58
##	ROE	-4.646452e-03	0.0037670512	-1.233445	2.174130e-01
##	NSI	-2.672897e-01	0.0769428186	-3.473875	5.132500e-04
##	NXF	-7.605044e-05	0.0000192296	-3.954864	7.663898e-05

```
# We can see taht Cluster SE's are different from SE's of original model.
#Most of SE's have increased, except the SE of NXF, which has decreased, meaning
#that it became more statistically significant.
```

ii)

```
EQ2_CL_PForm=felm(AvgMonRet~log(Size)+log(BTM)+ROE+NSI+NXF |0|0| PFormDate,
                  data = Signals)
SummaryEQ2_CL_PForm=summary(EQ2_CL_PForm)
SummaryEQ2_CL_PForm$coefficients
```

##		Estimate	Cluster s.e.	t value	Pr(> t)
##	(Intercept)	3.317109e+00	4.187847e-01	7.9207973	2.387718e-15
##	log(Size)	-2.543720e-01	5.797735e-02	-4.3874381	1.148288e-05
##	log(BTM)	3.171992e-01	1.191766e-01	2.6615897	7.778696e-03
##	ROE	-4.646452e-03	6.266680e-03	-0.7414535	4.584205e-01
##	NSI	-2.672897e-01	1.791278e-01	-1.4921738	1.356572e-01
##	NXF	-7.605044e-05	2.223206e-05	-3.4207554	6.247624e-04

```
SummaryEQ2$coefficients
```

```
##              Estimate   Std. Error   t value   Pr(>|t|)
## (Intercept)  3.317109e+00  0.0471461025  70.358067  0.000000e+00
## log(Size)    -2.543720e-01  0.0084849219 -29.979301  1.857746e-196
## log(BTM)     3.171992e-01  0.0197253279  16.080807  4.222881e-58
## ROE          -4.646452e-03  0.0037670512  -1.233445  2.174130e-01
## NSI          -2.672897e-01  0.0769428186  -3.473875  5.132500e-04
## NXF          -7.605044e-05  0.0000192296  -3.954864  7.663898e-05
```

*#This time Cluster SE's are substantially higher. As a result, both NSI and ROE
#became statistically insignificant with t-values of 0.74 and 1.49*

iii)

```
EQ2_CL_TwoWay=felm(AvgMonRet~log(Size)+log(BTM)+ROE+NSI+NXF | 0|0| PFormDate
                  +PermNo,
                  data = Signals)
SummaryEQ2_CL_TwoWay=summary(EQ2_CL_TwoWay)
SummaryEQ2_CL_TwoWay$coefficients
```

```
##              Estimate Cluster s.e.   t value   Pr(>|t|)
## (Intercept)  3.317109e+00  4.187397e-01  7.9216476  2.371455e-15
## log(Size)    -2.543720e-01  5.798881e-02 -4.3865709  1.152873e-05
## log(BTM)     3.171992e-01  1.190634e-01  2.6641193  7.720446e-03
## ROE          -4.646452e-03  5.724933e-03 -0.8116168  4.170138e-01
## NSI          -2.672897e-01  1.784762e-01 -1.4976215  1.342353e-01
## NXF          -7.605044e-05  2.706963e-05 -2.8094381  4.963917e-03
```

```
SummaryEQ2$coefficients
```

```
##              Estimate   Std. Error   t value   Pr(>|t|)
## (Intercept)  3.317109e+00  0.0471461025  70.358067  0.000000e+00
## log(Size)    -2.543720e-01  0.0084849219 -29.979301  1.857746e-196
## log(BTM)     3.171992e-01  0.0197253279  16.080807  4.222881e-58
## ROE          -4.646452e-03  0.0037670512  -1.233445  2.174130e-01
## NSI          -2.672897e-01  0.0769428186  -3.473875  5.132500e-04
## NXF          -7.605044e-05  0.0000192296  -3.954864  7.663898e-05
```

#Once again, SE's are larger and t-values are smaller, especially for ROE and NSI

Q4)

```
EQ2_CL_TwoWay_Fixed=felm(AvgMonRet~log(Size)+log(BTM)+ROE+NSI+NXF | PFormDate+PermNo | 0|PFormDate+
PermNo, data = Signals)
SummaryEQ2_CL_TwoWay_Fixed=summary(EQ2_CL_TwoWay_Fixed)
SummaryEQ2_CL_TwoWay_Fixed$coefficients
```



```
##               Estimate Cluster s.e.      t value      Pr(>|t|)
## log(Size) -1.782090e+00 1.109402e-01 -16.0635256 8.786273e-20
## log(BTM)  2.767890e-01 7.592581e-02  3.6455185 7.150836e-04
## ROE      -8.390773e-04 4.101249e-03 -0.2045907 8.388573e-01
## NSI      -2.348187e-01 1.265824e-01 -1.8550653 7.045116e-02
## NXF       5.742036e-06 1.589762e-05  0.3611884 7.197272e-01
```

SummaryEQ2\$coefficients

```
##               Estimate Std. Error      t value      Pr(>|t|)
## (Intercept)  3.317109e+00 0.0471461025  70.358067 0.000000e+00
## log(Size)    -2.543720e-01 0.0084849219 -29.979301 1.857746e-196
## log(BTM)     3.171992e-01 0.0197253279  16.080807 4.222881e-58
## ROE          -4.646452e-03 0.0037670512  -1.233445 2.174130e-01
## NSI          -2.672897e-01 0.0769428186  -3.473875 5.132500e-04
## NXF          -7.605044e-05 0.0000192296  -3.954864 7.663898e-05
```

We have added fixed effects for portfolio formation date and each firm. Because of that we experienced very different SE's, which are much larger than SE's that we have computed in Question 1 of Part B. Now, we can see that both ROE, NSI and NXF have large SE's and all three estimates are not significantly different from zero. In Question 1 of Part B, only ROE was insignificant.